ACIDS AND BASES 2

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Strength of acids and bases Acids (and bases) can be classified as **strong** or **weak** depending upon their behaviour when they dissolve in water. When these substances dissolve in water they ionise (break up and form charged particles known as ions). Figure 1 shows the general equation for the ionisation (dissociation) of an acid.

$$HA + H_2O \rightleftharpoons H_3O^+ + A^-$$

Figure 1

EXAMPLE

i **Give the equation for the ionisation of nitric acid.**

HNO_3 (l) + H_2O (l) $\rightleftharpoons$ H_3O^+ (aq) + NO_3^- (aq)

(HA) (A^-)

1 Give the equation for the ionisation of each acid.

a Hydrochloric acid, HCl (aq)

b Hydrobromic acid, HBr

c Sulfuric acid, H_2SO_4

The acids in Q1 are all strong acids. The molecules of these acids are completely (or close to 100%) ionised when they dissolve in water. There are certain acids that do not ionise very easily when they are dissolved in water. Only a small percentage (typically 1% to 5%) of the molecules break up to form ions. These are known as weak acids. Weak acids include ethanoic acid (acetic, CH_3COOH), methanoic (formic acid), phosphoric acid (H_3PO_4), citric acid ($C_6H_8O_7$) and hydrofluoric acid (HF).

2 Figure 2 shows the structural formula of methanoic acid. Give the molecular formula for methanoic acid.

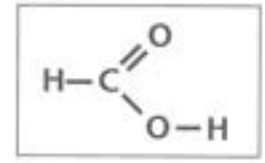


Figure 2

3 Show the ionisation products when each of these weak acids is dissolved in water.

a Ethanoic acid

b Hydrofluoric acid, HF

c Carbonic acid, H_2CO_3

Bases Common bases include metal hydrides, metal oxides and metal hydroxides. Strong bases (those that ionise completely) include sodium hydroxide (NaOH) and potassium hydroxide (KOH). Weak bases (those that only partially ionise) include ammonia (NH_3), ammonium hydroxide (NH_4OH) and methylamine (CH_3NH_2).

When methylamine is added to water, the following reaction occurs:

CH_3NH_2 (l) + H_2O (l) $\rightleftharpoons$ $CH_3NH_3^+$ (aq) + OH^- (aq)

4 Show the ionisation products when each base is dissolved in water.

a Sodium hydroxide

b Potassium hydroxide

c Ammonium

Equilibrium You may have noticed the equilibrium sign ($\rightleftharpoons$) in the equations showing the ionisation of acids and bases. This sign describes the balanced state at which forward reactions are equal to the reverse reactions (reactant to product and product back to reactant). For strong acids and bases, the reactions reach (or nearly reach) completion and at equilibrium are completely (or almost completely) ionised. For weak acids and bases the reaction does not go to completion. At equilibrium, weak acids and bases are mainly molecules with only a small proportion ionised.

5 Summarise the information about acids and bases on this page. Use the headings shown below.

Name	Formula	Acid/base	Strong/weak

6 Use the following terms to copy and complete the passage: *bases, completely, equilibrium, partially, strong*

Strong acids _____ ionise when they are dissolved in water. Weak acids only _____ ionise when they dissolve in water. _____ are the opposites of acids. There are _____ bases as well as weak ones. _____ describes a balanced state for a 'two way' reaction. This balanced state depends on the substances involved.

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Acid–base conjugate pairs When an acid reacts with a base, a conjugate acid and conjugate base are formed. **Conjugate** means *to join*. Conjugate acid–base pairs only differ by a proton (H^+). Figure 3 shows the reaction of ethanoic acid. Consider the reaction from left to right. Ethanoic (acetic) acid (CH_3COOH) has donated a proton to water (H_2O); hence ethanoic acid is an acid and water is a base. Looking at the reaction from right to left, the hydronium ion (H_3O^+) is an acid because it has donated a proton to the ethanoate (acetate) ion (CH_3COO^-). CH_3COOH and CH_3COO^- are conjugate acid–base pairs. The other conjugate acid–base pair is H_3O^+–H_2O.

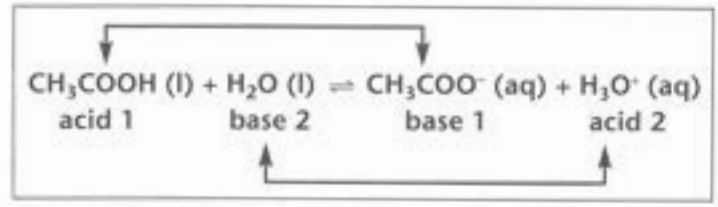


Figure 3

7 Refer to figure 3 to help you work out the conjugate acid–base pair for each reaction.

a $HNO_3\ (l) + H_2O\ (l) \rightleftharpoons H_3O^+\ (aq) + NO_3^-\ (aq)$

b $HBr\ (aq) + H_2O\ (l) \rightleftharpoons H_3O^+\ (aq) + Br^-\ (aq)$

c $H_2SO_4\ (aq) + H_2O\ (l) \rightleftharpoons 2H_3O^+\ (aq) + SO_4^{2-}\ (aq)$

d $C_6H_5NH_2\ (s) + H_2O\ (l) \rightleftharpoons C_6H_5NH_3^+\ (aq) + OH^-\ (aq)$

e $N_2H_4\ (l) + H_2O\ (l) \rightleftharpoons N_2H_5^+\ (aq) + OH^-\ (aq)$

f $NH_3\ (g) + H_2O\ (l) \rightleftharpoons NH_4^+\ (aq) + OH^-\ (aq)$

Amphiprotic substances Depending on their chemical environment, some species can donate and also (under different circumstances) accept protons. These are amphiprotic substances. Water forms hydroxide and hydronium ions (albeit in small quantities) as shown by the equation:

$$H_2O\ (l) + H_2O\ (l) \rightleftharpoons OH^-\ (aq) + H_3O^+\ (aq).$$

One water molecule has donated a proton to another water molecule.

The hydrogencarbonate (bicarbonate) ion has the formula HCO_3^- and only exists in aqueous solution. There is a similar term, **amphoteric**, which refers to substances that can react with either acids or bases.

8 In each reaction, state whether the hydrocarbonate ion is donating a proton or receiving a proton.

a $HCO_3^-\ (aq) + NH_4^+\ (aq) \rightleftharpoons H_2CO_3\ (aq) + NH_3\ (aq)$

b $HCO_3^-\ (aq) + H_2O\ (l) \rightleftharpoons H_3O^+\ (aq) + CO_3^{2-}\ (aq)$

Polyprotic acids Acids that generate one proton per molecule when they ionise are **monoprotic** acids. Some acids are capable of donating more than one proton per molecule. Sulfuric acid is a **diprotic** acid and phosphoric acid is a **triprotic** acid.

9 Study the equation. Explain what is meant by describing sulfuric acid as a *diprotic* acid.
$H_2SO_4\ (aq) + 2H_2O\ (l) \rightleftharpoons 2H_3O^+\ (aq) + SO_4^{2-}\ (aq)$

10 Study the equation. Explain what is meant by describing phosphoric acid as a *triprotic* acid.
$H_3PO_4\ (aq) + 3H_2O\ (l) \rightleftharpoons 3H_3O^+\ (aq) + PO_4^{3-}\ (aq)$

11 Mark each statement as true or false.

a Ions are electrically-neutral particles.

b Strong acids don't ionise completely when they dissolve in water.

c Group I metal hydroxides are often strong bases.

d The common (trivial) name for ethanoic acid is formic acid.

e In an acid-base conjugate pair, the two species only differ by a proton.

f Phosphoric acid forms salts called phosphates.

g All acids only donate one proton per molecule.

h Acids donate protons, bases accept them.

12 Define each term.

a Ionise

b Polyprotic acid

c Weak acids and strong acids

d Equilibrium

e Amphiprotic substances

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